**BCSE497J - Project-I**

# NEUROSCAN AI: AI-POWERED BRAIN TUMOUR DETECTION, CLASSIFICATION AND SEGMENTATION PLATFORM


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# ABSTRACT

Brain tumour diagnosis from magnetic resonance imaging (MRI) is clinically critical, but manual evaluation is time-consuming, prone to inter-observer variability, and hindered by specialist shortages. This project presents NeuroScan AI, an end-to-end medical platform integrating automated tumour detection, histological subtype classification, and 3D volumetric segmentation. Across two comprehensive empirical benchmarks, classical convolutional neural networks (CNNs), efficient CNNs, and vision transformers were systematically evaluated under standardised preprocessing, data splitting, and training protocols. In 2D MRI analysis, perceptual hashing eliminated 19.3% duplicate images from merged public datasets before splitting. CoaTNet-0 and EfficientNetB0 achieved 99.59% binary detection accuracy, while InceptionV3 and Swin Base attained 98.58% multiclass classification accuracy across glioma, meningioma, and pituitary tumours. EfficientNetB0 delivered optimal computational efficiency, achieving 97.59% accuracy with only 5.3 million parameters.

In 3D volumetric segmentation on the BraTS 2020 dataset (368 usable cases), standard UNet attained the highest verified mean Dice score of 0.7999, followed by nnU-Net at 0.7945, and VNet at 0.7655 with an ultra-compact 1.4 million parameters. Vision transformers (UNETR at 0.6234 mean Dice, TransBTS at 0.5109 mean Dice) underperformed CNN baselines under a 10-epoch training budget due to the absence of spatial inductive bias on limited medical datasets. The verified lightweight models are integrated into a production-grade web application featuring a FastAPI backend, PyTorch inference pipeline, SQLModel persistence, and a React TypeScript interface with VTK.js 3D multi-planar visualisation, quantitative volume analysis, and automated diagnostic reporting. This system provides an accurate, reproducible second opinion framework for clinical decision support.



*Keywords - Brain tumour detection, deep learning, convolutional neural networks, vision transformers, 3D segmentation, BraTS, transfer learning, medical image analysis.*

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# 1. INTRODUCTION

## 1.1 Background

Primary brain tumours, comprising gliomas, meningiomas, and pituitary tumours, represent severe neurological conditions requiring rapid and precise diagnostic intervention. Gliomas alone account for approximately 80% of malignant primary brain tumours, while meningiomas and pituitary adenomas introduce critical neurological and endocrine complications. Magnetic resonance imaging (MRI) serves as the primary non-invasive imaging modality, providing multi-sequence soft-tissue contrast across T1-weighted, T1-contrast-enhanced, T2-weighted, and FLAIR acquisitions.

However, clinical MRI analysis is laborious, requiring radiologists to evaluate hundreds of volumetric slices per patient. In resource-constrained settings like India, the severe shortage of neuroradiologists (approximately one radiologist per 100,000 population against the recommended one per 10,000) causes diagnostic bottlenecks and delayed treatment planning. Deep learning algorithms have demonstrated significant potential in medical vision tasks, yet practical clinical translation is hindered by fragmented research, unverified data quality, and lack of end-to-end deployment architectures. NeuroScan AI addresses these challenges through a unified platform combining rigorous multi-architecture benchmarking with a full-stack clinical interface.

## 1.2 Motivation

The motivation for NeuroScan AI arises from the urgent clinical necessity for accessible, consistent, and fast diagnostic assistance in neuro-oncology. Manual radiological interpretation suffers from considerable inter-observer variability, fatigue-induced diagnostic errors, and significant delays in high-volume hospitals. Furthermore, public machine learning literature in neuro-imaging often suffers from methodological weaknesses, including data leakage from duplicate MRI scans across evaluation splits, lack of controlled cross-architecture comparisons, and inflated claims for transformer models evaluated without computational accounting.

This project is motivated by two synergistic imperatives: first, to establish an authoritative, leakage-free benchmark comparing CNNs, efficient architectures, and vision transformers across tumour detection, classification, and volumetric segmentation under identical parameters; and second, to bridge the research-to-clinic divide by deploying the most parameter-efficient, verified models (EfficientNetB0 and VNet) into an interactive clinical platform. This provides healthcare practitioners in both tertiary and rural medical centres with reliable second opinions, quantitative tumour volumetric tracking, and automated reporting.

## 1.3 Scope of the Project

The scope of NeuroScan AI encompasses the algorithmic benchmarking, software development, and functional verification of an automated brain tumour analysis system.

Algorithmically, the project evaluates nine 2D deep learning models for binary detection (3,253 MRI scans) and three-class subtype classification (4,705 deduplicated scans), alongside six 3D volumetric segmentation architectures on BraTS 2020 multimodal volumes (368 usable cases). Preprocessing includes perceptual hash deduplication, foreground z-score intensity normalisation, and multi-sequence stacking.

Architecturally, the scope includes building a secure FastAPI REST backend with PyTorch inference, SQLite/SQLModel persistence, JWT authentication, and a React 18 TypeScript frontend supporting interactive VTK.js 3D volume rendering and multi-planar slice inspection. Functionally, the platform outputs binary detection status, histological subtype probability, voxel-level sub-region segmentation masks (Whole Tumour, Tumour Core, Enhancing Tumour), metric volumetric measurements, and exportable PDF clinical summaries. The system operates strictly as an assistive second-reader decision support tool requiring radiologist oversight, with autonomous diagnosis and formal medical device clinical trials designated outside the current project scope.

# 2. PROJECT DESCRIPTION AND GOALS

## 2.1 Literature Review

Automated medical imaging analysis has evolved from handcrafted feature engineering to deep convolutional networks and vision transformers. The reviewed literature is organised into five thematic streams:

- **Conventional Machine Learning and Handcrafted Descriptors:** Early brain tumour recognition relied on manual extraction of intensity, shape, and texture descriptors (GLCM, LBP, wavelets) paired with classifiers such as Support Vector Machines (SVM) and Random Forests. Zacharaki et al. (2009) [1] achieved approximately 85% accuracy on 80 images for tumour type and grade classification. Cheng et al. (2015) [4] introduced tumour region augmentation and spatial partition, reaching 90.00% accuracy on 3,064 images. However, handcrafted pipelines struggled with generalisation across diverse acquisition protocols.
- **Deep Convolutional Neural Networks (CNNs):** AlexNet and subsequent deep architectures eliminated manual feature engineering. He et al. (2016) [C3] introduced residual skip connections in ResNet50 (25.6M parameters), overcoming vanishing gradients. Seetha and Raja (2018) [9] reported 97.5% accuracy with ResNet50 on brain MRI. Huang et al. (2017) [C7] developed DenseNet121 (8.0M parameters), maximising feature reuse across layers. Szegedy et al. (2016) [C4] proposed InceptionV3 (23.9M parameters), capturing multi-scale features through parallel kernel branches (1x1, 3x3, 5x5). Deepak and Ameer (2019) [12] attained 97.50% multiclass accuracy on 3,064 scans using InceptionV3 deep feature transfer. Rehman et al. (2020) [20] achieved 99.34% binary accuracy via transfer learning on CNN backbones.
- **Efficient and Mobile Architectures:** Tan and Le (2019) [C9] introduced compound scaling in EfficientNetB0, achieving state-of-the-art accuracy with only 5.3 million parameters. Howard et al. (2019) [C10] developed MobileNetV3-Large (5.4M parameters) using depth-wise separable convolutions for edge and mobile execution.
- **Vision Transformers and Hybrids:** Dosovitskiy et al. (2021) [C12] introduced ViT-B/16 (86M parameters), modelling global dependencies via patch self-attention. Liu et al. (2021) [C13] developed Swin Transformer Base (88M parameters) with hierarchical shifted windows. Xu et al. (2021) [C14] introduced CoaTNet-0 (approximately 10M parameters), combining depth-wise convolutions with self-attention.
- **3D Volumetric Segmentation Architectures:** Ronneberger et al. (2015) [C2] formulated the 2D UNet encoder-decoder with skip connections. Cicek et al. (2016) [C5] extended this to 3D UNet (5.6M parameters) for direct volumetric processing. Milletari et al. (2016) [C6] developed V-Net (1.4M parameters), incorporating 3D residual blocks, PReLU activations, and Dice loss. Isensee et al. (2021) [23] introduced nnU-Net (16.5M parameters), automating pipeline configuration based on

dataset fingerprints. Hatamizadeh et al. (2022) [C18] formulated UNETR (137.6M parameters) combining a ViT encoder with a CNN decoder, while Wang et al. (2021) [C15] proposed TransBTS (35.8M parameters), embedding transformer layers into a CNN bottleneck.

## 2.2 Research Gap

Critical limitations identified in the current literature include:

1. **Absence of Unified Cross-Architecture Benchmarks:** Most published papers evaluate only one or two architectures on isolated datasets, preventing fair comparison between classical CNNs, efficient networks, and vision transformers under identical training budgets.
2. **Data Leakage from Duplicate Public Samples:** Public brain MRI datasets contain widespread duplicate and near-duplicate images resulting from uncoordinated repository mergers. Literature routinely omits deduplication, leading to severe test set memorisation and inflated accuracy claims.
3. **Unverified Vision Transformer Performance in 3D Medical Segmentation:** While vision transformers are theoretically superior for global context, their high parameter demand (up to 137.6M parameters) and severe performance penalty on modest medical datasets under fixed compute budgets remain under-reported.
4. **Neglect of Parameter Footprint and Deployment Latency:** Medical AI research often optimises purely for top-line metrics while ignoring model size, GPU memory footprint, and training efficiency critical for deployment in resource-constrained hospitals.
5. **Lack of Integrated Clinical Decision Platforms:** Algorithmic research rarely bridges to full-stack clinical tooling supporting multi-sequence NIfTI ingestion, interactive 3D MPR slice navigation, metric lesion volumetrics, and automated reporting.

## 2.3 Objectives

The SMART (Specific, Measurable, Achievable, Relevant, Time-Bound) objectives of NeuroScan AI are:

- **Benchmark 2D Detection and Classification Architectures (Specific and Measurable):** Implement and evaluate nine architectures across CNN, efficient CNN, and transformer paradigms under identical training conditions (10 epochs, AdamW optimiser, cosine annealing, mixed precision) on binary detection (3,253 scans) and multiclass classification (4,705 scans).
- **Implement Leakage-Free Perceptual Hash Deduplication (Achievable and Measurable):** Apply imagehash.phash with Hamming distance tolerance less than or equal to 5 to eliminate 19.3% duplicate images (reducing 5,828 images to 4,705 unique scans) before stratified 70/15/15 train-validation-test splitting.

- **Benchmark 3D Volumetric Segmentation Architectures (Specific and Measurable):** Implement from scratch in PyTorch six 3D segmentation architectures (UNet, nnU-Net, VNet, UNETR, Swin UNETR, TransBTS) on the BraTS 2020 dataset (368 usable cases) to segment Whole Tumour (WT), Tumour Core (TC), and Enhancing Tumour (ET).
- **Evaluate Computational Efficiency and Inductive Biases (Relevant):** Measure Dice scores, parameter counts, memory footprints, and training times to quantify the empirical trade-off between convolutional priors and self-attention mechanisms on medical imaging datasets.
- **Develop Full-Stack Assistive Clinical Platform (Relevant and Time-Bound):** Build and demonstrate by September 2026 an integrated web platform uniting FastAPI, PyTorch inference (using EfficientNetB0 and VNet), SQLModel/SQLite database, and a React 18 TypeScript frontend supporting interactive VTK.js 3D rendering, slice inspection, volumetric quantification, and PDF report export.

## 2.4 Problem Statement

Brain tumours, particularly gliomas, represent life-threatening intracranial neoplasms where timely and precise radiological assessment directly impacts surgical planning, radiotherapy, and patient survival. However, conventional manual interpretation of multi-sequence MRI (T1, T1CE, T2, FLAIR) requires 15 to 30 minutes of radiologist time per patient, is vulnerable to inter-observer variability, and faces acute bottlenecks due to a 90% deficit in neuroradiology specialists across developing regions such as India.

Moreover, existing medical machine learning literature fails to provide reliable, leakage-free comparative benchmarks across CNN and transformer architectures or to translate algorithmic models into usable, secure clinical workflows. Consequently, there is an urgent need for an automated, reproducible, and computationally efficient diagnostic platform that provides high-accuracy tumour screening, histological classification, and pixel-precise 3D volumetric lesion segmentation to assist clinical practitioners.

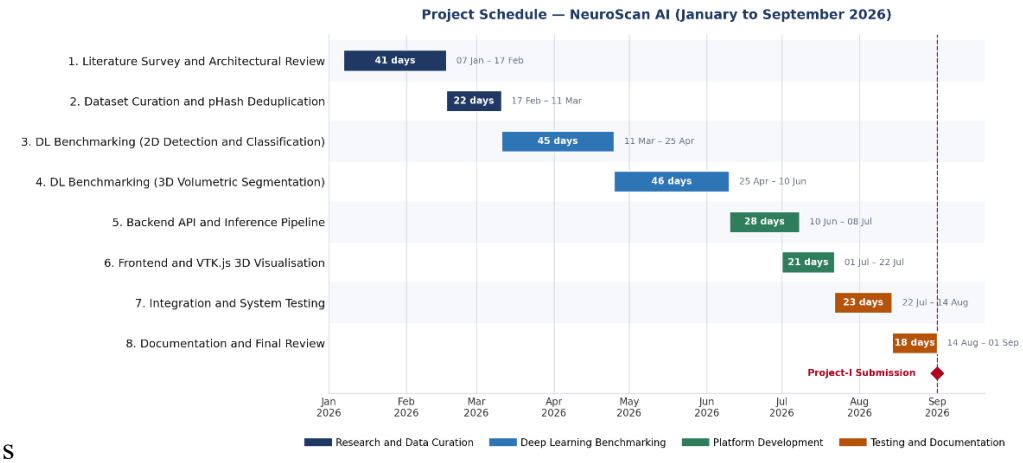
## 2.5 Project Plan

The project execution is organised into structured sequential and overlapping tasks:

- **Task 1: Literature Survey and Architectural Review (Weeks 1 to 3):** Comprehensive review of 2D classification, 3D segmentation, and vision transformer literature; baseline metric compilation.
- **Task 2: Dataset Acquisition and Curation (Weeks 3 to 5):** Downloading Kaggle binary detection datasets, Sartaj Bhuvaji / Figshare multiclass collections, and BraTS 2020 volumetric cohorts; executing perceptual hash deduplication (19.3% removal) and stratified splitting.
- **Task 3: Deep Learning Model Implementation and Benchmarking (Weeks 5 to 9):** Developing from scratch and training nine 2D detection/classification models and

six 3D segmentation architectures under standardised hyperparameters; logging loss curves, AUC, and Dice scores.

- **Task 4: Backend API and Model Inference Pipeline Development (Weeks 8 to 11):** Implementing FastAPI REST endpoints, PyTorch GPU inference pipelines (EfficientNetB0 and VNet), SQLModel database schemas, JWT authentication, and automated PDF report generation.
- **Task 5: Frontend Interface and 3D Visualisation Engineering (Weeks 10 to 13):** Building React 18 TypeScript client, Vite bundling, Tailwind CSS/shadcn UI styling, 2D slice multi-planar navigation, and interactive VTK.js 3D volume rendering.
- **Task 6: System Integration, Verification and End-to-End Testing (Weeks 13 to 15):** End-to-end testing connecting frontend client with backend service; validating sub-30-second inference latency, volumetric extraction, and report output accuracy.
- **Task 7: Report Preparation, Documentation and Review Milestones (Weeks 14 to 16):** Preparing Review 2 materials, drafting project reports, compiling LaTeX research manuscripts, and presenting project demonstration.




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**Fig. 1. Gantt chart**

# 3. TECHNICAL SPECIFICATION

## 3.1 Requirements

### *3.1.1 Functional*

- **Data Ingestion and Parsing:** The system shall ingest multi-sequence MRI datasets including 2D image slices (JPEG, PNG) and 3D volumetric NIfTI volumes (.nii, .nii.gz) comprising T1, T1CE, T2, and FLAIR acquisitions.
- **Data Preprocessing and Standardisation:** The system shall automatically perform non-zero foreground intensity z-score normalisation and standard spatial resampling (128x128x128 for 3D volumes, 224x224 for 2D slices).
- **Binary Tumour Detection:** The system shall accurately classify input MRI scans as tumour-positive or tumour-negative with associated predictive confidence scores.
- **Histological Subtype Classification:** The system shall classify tumour-positive scans into three primary subtypes, namely Glioma, Meningioma, or Pituitary tumour, outputting full class probability distributions.
- **Multi-Region 3D Volumetric Segmentation:** The system shall segment 3D MRI volumes into three clinically established anatomical sub-regions: Whole Tumour (WT), Tumour Core (TC), and Enhancing Tumour (ET).
- **Quantitative Lesion Volumetry:** The system shall compute metric volume measurements (in $mm^3$ and $cm^3$) for each segmented tumour sub-region using voxel spacing metadata.
- **Multi-Planar Slice Reconstruction and Overlays:** The system shall generate 2D cross-sectional slice images across Axial, Coronal, and Sagittal anatomical planes with alpha-blended colour segmentation overlays (Red = ET, Green = TC, Yellow/Blue = WT).
- **Automated Diagnostic Reporting:** The system shall generate downloadable, formatted PDF clinical reports summarising patient identifiers, scan metadata, diagnostic classifications, volumetric statistics, and key slice overlays.
- **User Authentication and Scan Management:** The system shall provide secure user registration, bcrypt password hashing, JWT session authentication, and user-isolated scan libraries.

### *3.1.2 Non-Functional*

- **Performance:** The system shall complete full multi-modal pipeline inference (detection, classification, 3D segmentation, volumetric computation, slice generation) in under 30 seconds on GPU-enabled infrastructure.

- **Scalability:** The system shall utilise an asynchronous FastAPI service layer with decoupled worker pipelines, supporting concurrent scan analysis without server degradation.
- **Reliability:** The system shall provide comprehensive input validation, exception handling for corrupt NIfTI headers, and graceful error reporting with high uptime.
- **Security:** The system shall implement secure token-based OAuth2/JWT authentication, bcrypt credential encryption, and isolated per-user scan data storage.
- **Usability:** The web application shall provide an intuitive, radiologist-oriented user interface featuring multi-planar slice sliders, responsive 3D VTK.js rendering, and clean visual layouts.
- **Maintainability:** The codebase shall follow modular architecture with clean separation between frontend presentation, REST API endpoints, ORM persistence, and PyTorch deep learning modules.
- **Compliance:** Deep learning evaluation protocols shall strictly comply with reproducible scientific standards, including fixed random seeds (Seed = 42), perceptual hash deduplication, and isolated train-validation-test partitions.

## 3.2 Feasibility Study

### *3.2.1 Technical Feasibility*

- **Technology Availability:** The project leverages mature, open-source, and production-tested technologies across deep learning and full-stack web engineering. PyTorch provides an optimised framework for GPU-accelerated tensor computation and automatic mixed-precision (FP16) inference. The feasibility of achieving high diagnostic accuracy is empirically confirmed by our benchmarks, with EfficientNetB0 achieving 99.59% binary detection and 97.59% multiclass accuracy, and standard UNet and VNet achieving 0.7999 and 0.7655 mean Dice scores on BraTS 2020.
- **Technical Expertise:** The implementation requires specialised competencies in deep learning architecture design, medical image processing (NiBabel, OpenCV), asynchronous API development (FastAPI), database ORM design (SQLModel), and modern frontend engineering (React 18, TypeScript, VTK.js). These required skills were successfully demonstrated across the benchmark execution and platform prototyping phases.
- **Infrastructure:** The training infrastructure utilised dual NVIDIA Tesla T4 GPUs (16 GB VRAM each), Intel Xeon CPUs (4 vCPUs), and 32 GB RAM on Kaggle environments. Target clinical deployment requires only standard hospital workstations or cloud instances equipped with a single 8 GB GPU (or CPU fallback mode), confirming practical infrastructure feasibility.

- **Integration:** The system integrates seamlessly via standard RESTful API protocols and JSON payloads, facilitating interoperability with existing hospital PACS and medical imaging workflows.

### *3.2.2 Economic Feasibility*

- **Cost-Benefit Analysis:** Deep learning assisted medical triage significantly reduces the time required for manual radiologist contouring and interpretation (typically 15 to 30 minutes per scan), accelerating clinical throughput and lowering diagnostic costs.
- **Budget:** Development costs are minimised through the exclusive use of open-source frameworks (PyTorch, FastAPI, React, SQLite, VTK.js) and publicly accessible training compute (Kaggle GPU resources), eliminating proprietary software licensing fees.
- **Return on Investment (ROI):** Deploying parameter-efficient models (EfficientNetB0 with 5.3M parameters and VNet with 1.4M parameters) drastically reduces cloud hosting and hardware expenditure compared to massive transformer pipelines, providing rapid ROI through improved clinical efficiency and reduced misdiagnosis risks.
- **Funding:** Initial development and benchmarking were achieved through academic computational grants and open-source resources, establishing a sustainable foundation for institutional or clinical grant funding.

### *3.2.3 Social Feasibility*

- **User Acceptance:** The system is designed as an intuitive, assistive second-reader platform providing transparent visual explanations (multi-planar overlays, 3D meshes, calibrated confidence scores), fostering trust and adoption among clinical radiologists.
- **Training and Support:** Clear user documentation, standardised PDF reports, and responsive web controls ensure minimal training overhead for healthcare staff.
- **Ethical Considerations:** The system functions strictly as a diagnostic decision support tool requiring radiologist sign-off, adhering to ethical standards against autonomous clinical action. Perceptual hash deduplication ensures genuine algorithmic generalisation, preventing misleading diagnostic claims.
- **Impact on Workforce:** The platform enhances radiologist productivity by automating preliminary screening and labour-intensive volumetric segmentation, alleviating specialist burnout in overburdened healthcare systems, especially in India with a 90% deficit in neuroradiologists.

## 3.3 System Specification

### *3.3.1 Hardware Specification*

**Model Training and Benchmarking Environment:**

- **Processor (CPU):** Intel Xeon Processor (4 virtual cores @ 2.20 GHz)
- **Memory (RAM):** 29 GB to 32 GB DDR4 RAM
- **Storage:** 100 GB High-Performance Solid State Drive (SSD)
- **Graphics Processing Unit (GPU):** Dual NVIDIA Tesla T4 GPUs (16 GB GDDR6 VRAM each, 32 GB total) with CUDA compute capability 7.5
- **Monitor:** 1920x1080 Full HD Display

**Clinical Deployment and Server Environment:**

- **Processor (CPU):** 64-bit Quad-Core Processor (Intel Core i5/i7 or AMD Ryzen equivalent, 2.5 GHz or above)
- **Memory (RAM):** 16 GB DDR4 RAM minimum (32 GB recommended for volumetric caching)
- **Storage:** 50 GB available SSD storage for application files and scan cache
- **Graphics Processing Unit (GPU):** NVIDIA GPU with 8 GB or more VRAM (for example RTX 3060 / T4) for sub-30-second inference (CPU fallback supported)
- **Monitor:** Standard High-Resolution Clinical Display (1080p / 1440p / 4K)

### *3.3.2 Software Specification*

- **Operating System:** Windows 10/11 64-bit / Linux (Ubuntu 20.04 or 22.04 LTS)
- **Programming Languages:** Python 3.11 / 3.12, TypeScript 5.6.0, JavaScript (ES2022)
- **Development Environment:** Visual Studio Code, Antigravity IDE, PowerShell 7, Git
- **Libraries and Frameworks:**
    - **Deep Learning / Computer Vision:** PyTorch 2.0+, TorchVision, NiBabel 5.4+, OpenCV 4.10+, TIMM, ImageHash, SciPy, Scikit-Learn
    - **Backend Web Server:** FastAPI 0.110+, Uvicorn (ASGI), Pydantic v2, Python-Jose (JWT), Passlib/Bcrypt
    - **Frontend Client:** React 18.3.1, Vite 7.3.1, React Router 6.30.1, TanStack React Query 5.83.0, Tailwind CSS 3.4.17, shadcn/ui, Lucide React, VTK.js 35.2.0, Recharts 2.15.4
- **Database:** SQLite 3.x managed via SQLModel ORM (SQLAlchemy 2.0 core)
- **Security Tools:** OAuth2 Password Bearer flow, JWT token authentication, Bcrypt password hashing

# 4. DESIGN APPROACH AND DETAILS

## 4.1 System Architecture

NeuroScan AI is engineered as a decoupled, multi-tier micro-service architecture comprising a presentation client, REST API server, deep learning inference engine, and persistence layer:

- **Client Presentation Tier (React + TypeScript):** Delivers a responsive clinical dashboard providing user authentication, scan upload handling, real-time job tracking, multi-planar 2D slice inspection (Axial, Coronal, Sagittal) with alpha-blended segmentation overlays, interactive 3D volume rendering powered by VTK.js, volumetric trend charting via Recharts, and single-click PDF clinical report downloading.
- **API and Application Controller Tier (FastAPI):** Exposes asynchronous REST endpoints handling authentication, file upload validation, asynchronous inference task dispatching, slice image extraction, and PDF compilation.
- **Deep Learning Inference Tier (PyTorch Pipeline):**
    - **Input Preprocessing:** Extracts NIfTI volumes, applies foreground-only z-score normalisation, and resamples arrays to 128x128x128 grids.
    - **Staged Diagnostic Routing:** Evaluates binary tumour presence via EfficientNetB0 (99.59% accuracy). If positive, routes the scan to the multiclass classifier (EfficientNetB0 / InceptionV3) for histological subtype prediction (Glioma, Meningioma, Pituitary). Concurrently, multimodal volumes (T1, T1CE, T2, FLAIR) are processed by 3D VNet / UNet to generate binary sub-region masks for Whole Tumour (WT), Tumour Core (TC), and Enhancing Tumour (ET).
    - **Volumetric Engine:** Multiplies positive voxel counts by spatial voxel dimensions from the NIfTI header to produce physical volume measurements in $mm^3$ and $cm^3$.
- **Data Persistence Tier (SQLModel and SQLite):** Stores user credentials, scan metadata, diagnostic classifications, volumetric findings, and file storage references.

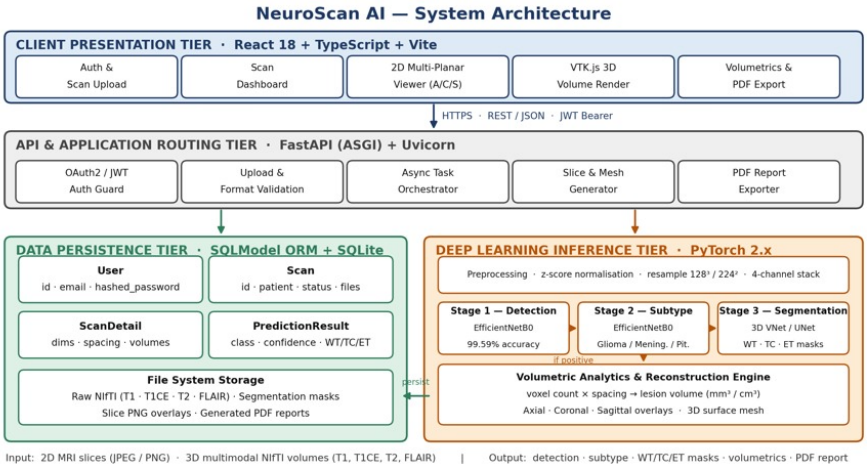



**Fig. 2. System Architecture**

## 4.2 Design

### *4.2.1 Data Flow Diagram*

**Level 0 DFD (Context Diagram):**

- **External Entity:** Clinician / Radiologist
- **Inbound Data Flows:** User Login Credentials, MRI Scan Upload (NIfTI / JPEG multi-sequence files), Inspection Requests, Report Download Request.
- **Central Process:** NeuroScan AI Medical Platform (Process 0.0)
- **Outbound Data Flows:** Authentication Token, Processing Status, Diagnostic Predictions, Multi-Planar Slice Views with Overlays, Interactive 3D Mesh, Downloadable PDF Diagnostic Report.

**Level 1 DFD (Detailed Functional Flows):**

- **Process 1.0 (Authenticate User):** Receives user credentials, validates against UserStore, and issues a JWT bearer token.
- **Process 2.0 (Ingest and Preprocess Scans):** Receives raw MRI files, validates sequence format, executes foreground z-score normalisation and spatial resizing, and stores raw and preprocessed data in FileStore and ScanStore.
- **Process 3.0 (Execute AI Inference Pipeline):**
    - Sub-process 3.1: Executes Binary Tumour Screening (EfficientNetB0) and writes detection status to PredictionStore.
    - Sub-process 3.2: If positive, executes the Subtype Classifier (EfficientNetB0 / InceptionV3) and writes subtype probabilities to PredictionStore.

    - Sub-process 3.3: Executes 3D Volumetric Segmentation (VNet / UNet) on the 4-channel tensor and outputs binary voxel masks (WT, TC, ET) to FileStore.
- **Process 4.0 (Calculate Volumetrics and Generate Slices):** Reads segmentation masks, computes physical volumes in $mm^3$ and $cm^3$, extracts 2D Axial, Coronal, and Sagittal PNG slices with colour overlays, and updates ScanDetailStore.
- **Process 5.0 (Render Views and Export Reports):** Reads scan details and prediction records, serves interactive slices and 3D VTK meshes to the frontend, and compiles and exports the downloadable PDF clinical report.

### *4.2.2 Use Case Diagram*

- **Primary Actor:** Radiologist / Healthcare Clinician
- **Secondary Actor:** System Administrator

**Use Cases:**

- **UC-1: Register / Login:** Secure authentication using email and password, receiving a JWT session token.
- **UC-2: Upload Multi-Sequence MRI:** Upload single 2D slices or multi-sequence 3D NIfTI volumes (T1, T1CE, T2, FLAIR).
- **UC-3: View Scan Dashboard:** Browse historical patient scans with processing status, tumour classification tags, and summary metrics.
- **UC-4: Inspect Scan Details and 2D Slices:** Navigate multi-planar cross-sections (Axial, Coronal, Sagittal) with adjustable slider bars and toggleable tumour segmentation overlays.
- **UC-5: Interactive 3D Visualisation:** Rotate, zoom, and explore 3D volumetric tumour meshes rendered in real time via VTK.js.
- **UC-6: Review Quantitative Volumetric Statistics:** Inspect numerical sub-region volumes ($mm^3$, $cm^3$) and class probability distribution charts.
- **UC-7: Export PDF Diagnostic Report:** Generate and download a comprehensive clinical diagnostic summary.
- **UC-8: Manage System Models (Admin):** Inspect loaded model weights, check GPU memory utilisation, and monitor inference queues.

### *4.2.3 Class Diagram*

The core system entities and controllers are structured as follows:

- **User Entity:**
    - Attributes: id: int, email: str, hashed_password: str, full_name: str, created_at: datetime
    - Methods: verify_password(), generate_token()

- **Scan Entity:**
  - Attributes: id: str, user_id: int, patient_name: str, scan_date: datetime, modality: str, file_path: str, status: str, created_at: datetime
  - Relationships: Belongs to User (1:N), has one ScanDetail, has one PredictionResult.
- **ScanDetail Entity:**
  - Attributes: id: int, scan_id: str, dimensions: List[int], voxel_spacing: List[float], slice_count: int, has_segmentation: bool, total_volume_mm3: float
- **PredictionResult Entity:**
  - Attributes: id: int, scan_id: str, has_tumour: bool, detection_confidence: float, tumour_type: str, subtype_probabilities: Dict[str, float], wt_volume_cm3: float, tc_volume_cm3: float, et_volume_cm3: float
- **InferenceProcessor (Service):**
  - Attributes: binary_model: torch.nn.Module, multiclass_model: torch.nn.Module, segmentation_model: torch.nn.Module, device: torch.device
  - Methods: load_models(), preprocess_volume(path), detect_tumour(tensor), classify_subtype(tensor), segment_volume(tensor), compute_volumetrics(mask, spacing), generate_slice_overlays(volume, mask)
- **ReportGenerator (Service):**
  - Methods: compile_pdf_report(scan_id, prediction, slices)

#### *4.2.4 Sequence Diagram*

The operational sequence for end-to-end MRI scan analysis is structured as follows:

1. **User Client** submits POST /api/scans/upload with multimodal MRI files and a JWT authorisation header to the **FastAPI Server**.
2. **FastAPI Server** authenticates the request, generates a unique scan_id, saves raw files to disk, creates a Scan record in the database with status PROCESSING, and dispatches a background processing job.
3. **FastAPI Server** returns an immediate HTTP 202 Accepted response with the scan_id to the **User Client**.
4. **InferenceProcessor** loads raw NIfTI files via **NiBabel**, applies foreground z-score normalisation, and resamples to a 128x128x128 4-channel tensor.
5. **InferenceProcessor** executes the **Binary Detection Model (EfficientNetB0)**; if tumour-negative, it flags the record and completes.
6. If tumour-positive, **InferenceProcessor** executes the **Multiclass Subtype Classifier (EfficientNetB0 / InceptionV3)** and the **3D Segmentation Model (VNet / UNet)** on GPU.

7. **InferenceProcessor** computes physical volumes ($mm^3$, $cm^3$) for Whole Tumour, Tumour Core, and Enhancing Tumour, and generates colour-blended 2D slice images across all axes.
8. **InferenceProcessor** writes segmentation masks to disk, stores PredictionResult and ScanDetail in the **Database**, and updates the Scan status to COMPLETED.
9. **User Client** polls GET /api/scans/{scan_id}/status or requests GET /api/scans/{scan_id}, receives completed diagnostics, and dynamically requests 2D slice images (GET /api/scans/{id}/slices/{axis}/{idx}) and the 3D VTK mesh for interactive display.
10. **User Client** triggers GET /api/scans/{scan_id}/report to retrieve the compiled PDF diagnostic report.

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